

Proof. Consider the quadratic

$$f(t) = \sum_{i=1}^n (a_i - tb_i)^2 \geq 0 \quad (t \in \mathbb{R}).$$

Expanding we obtain

$$\begin{aligned} f(t) &= \sum_{i=1}^n (a_i^2 - 2ta_i b_i + t^2 b_i^2) \\ &= \sum_{i=1}^n a_i^2 - 2t \sum_{i=1}^n a_i b_i + t^2 \sum_{i=1}^n b_i^2 \\ &= C + Bt + At^2, \end{aligned}$$

where

$$A = \sum_{i=1}^n b_i^2, \quad B = -2 \sum_{i=1}^n a_i b_i, \quad C = \sum_{i=1}^n a_i^2.$$

If $A > 0$ (i.e. not all b_i are zero), the quadratic $At^2 + Bt + C$ is non-negative for every real t , hence its discriminant must be non-positive:

$$B^2 - 4AC \leq 0.$$

If $A = 0$ then $b_i = 0$ for all i , and the inequality to be proved reduces to $0 \leq 0$, which is true.

Substituting the expressions for A, B, C into the discriminant inequality yields

$$\begin{aligned} \left(-2 \sum_{i=1}^n a_i b_i\right)^2 - 4 \left(\sum_{i=1}^n b_i^2\right) \left(\sum_{i=1}^n a_i^2\right) &\leq 0, \\ \left(\sum_{i=1}^n a_i b_i\right)^2 &\leq \left(\sum_{i=1}^n a_i^2\right) \left(\sum_{i=1}^n b_i^2\right), \end{aligned}$$

which is the Cauchy–Schwarz inequality.

Equality condition. Equality holds exactly when the discriminant is zero. For $A > 0$ this means that $f(t)$ has a double root

$$t_0 = -\frac{B}{2A} = \frac{\sum_{i=1}^n a_i b_i}{\sum_{i=1}^n b_i^2}.$$

Since $f(t_0) = 0$ and each term $(a_i - t_0 b_i)^2$ is non-negative, every term must vanish; thus $a_i = t_0 b_i$ for all i .

If $A = 0$ (all $b_i = 0$), the inequality is an equality for any a_i , which is also compatible with $a_i = \lambda b_i$ for arbitrary λ .

Hence equality holds precisely when there exists $\lambda \in \mathbb{R}$ such that $a_i = \lambda b_i$ for every i . $\square$